Code No: 114CQ **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech II Year II Semester Examinations, April - 2018
DATABASE MANAGEMENT SYSTEMS
(Common to CSE, IT)

Time: 3 Hours **Max. Marks: 75**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A.
Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

PART- A

(25 Marks)

1.a) What are five main functions of database Admistrator? [2]
b) List four significant difference between a file processing system and DBMS. [3]
c) Explain about set membership in SQL. [2]
d) Explain Outer join operators in relational algebra. [3]
e) What is functional dependency? [2]
f) Explain about join Dependency. [3]
g) What is fuzzy check point? [2]
h) What are the different types of log records and when they are written? [3]
i) What is the main difference between static dynamic hashing? [2]
j) What are the main difference between static and dynamic index structure? [3]

PART-B

(50 Marks)

2.a) What is data model? What is the relational data model? What is data independence and how does a DBMS support it.
b) What is E-R Model? Draw an E-R Diagram for online book store. [5+5]

OR

3.a) Explain the difference between external, internal and conceptual schemas. How these different schema layers are related to logical and physical data independence.
b) Draw an ER-Diagram for Specialization, generalization and aggregation. [5+5]

4.a) Explain Domain Integrity constraints in SQL.
b) Give an overview of active databases. [5+5]

OR

5.a) Explain various DML statement in SQL with example.
b) Explain Aggregate and database modification operators in relational algebra. [5+5]

6.a) What is normalization? What are the conditions are required for a relation to be in 1NF, 2NF and 3NF. Explain with examples.
b) Explain the following with examples:
i) Join Dependency ii) multivalued dependency. [5+5]

OR

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7.a) Suppose that we decompose the schema $R = (A, B, C, D, E)$ into R1 (A, B, C) and R2 (A, D, E). Show that this decomposition is lossless-join decomposition if the following set F of functional dependencies holds:
$A \rightarrow BC,\ CD \rightarrow E, B \rightarrow D, E \rightarrow A$

b) What are the condition are required for 4NF and 5NF Normal forms explain with examples? [5+5]

8.a) What is Concurrency control? Explain how multiple granularity protocol is used to Control concurrent transactions.

b) Explain what happens if there are crashes during undo phase of recovery? What if there are crashes during the analysis phase and redo phase. [5+5]

OR

9.a) Explain conservative *Time Stamp-Based Concurrency* Control protocol.

b) What are the problems with lock-based concurrency control protocols. [5+5]

10.a) Describe how insert and delete operations work in ISAM.

b) What are the problems with storage of data on external storage? Explain. [5+5]

OR

11.a) Describe the B+ tree insert and delete operations with examples.

b) What is *Hashing*? Discuss the relation between extendable hashing and Linear Hashing. What are their relative merits? [5+5]

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